

Chibuike Robinson Umeike

AI/ML Research Scientist — Multimodal Foundation Models, LLM, & Scientific AI

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PROFESSIONAL SUMMARY

AI/ML Research Scientist specializing in foundation model adaptation, generative AI for scientific applications (GenAI4Science), and large-scale multimodal deep learning. At Argonne National Laboratory, engineered domain-aware vision-language models using reinforcement learning (GRPO/PPO) for structured scientific image interpretation, built end-to-end data pipelines processing 165K+ artifacts from **42,673 scientific articles**, and fine-tuned large VLMs achieving **135% performance improvement** and **96.6% hallucination reduction** over baselines. Developed **PtychoBench** for evaluating foundation model adaptation to X-ray ptychography data from the Advanced Photon Source, and integrated generative AI into production scientific applications. Experienced in multi-GPU training, scalable data pipelines, and reproducible research on leadership-class HPC systems. **8 first-author papers** at NeurIPS, IEEE ISBI, and IEEE venues.

TECHNICAL SKILLS

- **Generative AI:** GRPO/PPO alignment, RLHF, reinforcement learning for structured generation, LLM-as-judge Eval, OpenAI (Gpt-4o)/Anthropic Claude API (claude-sonnet-4-20250514), instruction tuning, structured output generation
- **Foundation Models & VLMs:** LLaVA, Qwen-2.5VL, Llama-3.2-Vision, InternVL, Vicuna, CLIP – fine-tuning (LoRA), inference optimization, vLLM, Hugging Face Transformers
- **Computer Vision:** YOLO, RT-DETR, ResNet, ConvNeXt, SAM2, ViT, EfficientNet – object detection, segmentation, classification, OOD generalization, ordinal supervision
- **Scientific Data:** X-ray ptychography (synchrotron/APS), biomedical imaging, heterogeneous document parsing, figure-caption extraction, structured scientific data curation (171K+ samples)
- **ML Frameworks:** PyTorch, TensorFlow, OpenCV, Scikit-Learn, VERL, Weights & Biases
- **Production & HPC:** Multi-GPU distributed training, ALCF HPC, real-time inference pipelines, scalable data curation
- **Languages & Tools:** Python, C++, C#, Java, MATLAB, Bash, Git, PyTorch, Keras, TensorFlow, Numpy, Unsloth

PROFESSIONAL EXPERIENCE

- **Argonne National Laboratory (U.S. Department of Energy)** Lemont, IL
 - **Research Technical Aide** — *AI/ML for Scientific Computing* May 2024 – Aug 2025
 - **Foundation Models for Scientific Imaging (Summer 2025):** • Engineered domain-aware vision-language models (Qwen-2.5VL 3B/7B/72B, Llama-3.2-Vision 11B/90B) using reinforcement learning (GRPO/PPO) for structured interpretation of X-ray ptychography data from the Advanced Photon Source. • Designed and released PtychoBench, a benchmark for evaluating foundation model adaptation in low-data physical science regimes. • Managed end-to-end pipeline from raw synchrotron data ingestion to structured model-ready representations, integrating domain expert feedback. • *Outcome: First-author paper at NeurIPS 2025 AI4MAT Workshop; NeurIPS Travel Grant recipient.*
 - **Multimodal Knowledge Extraction from Scientific Literature (Summer 2024):** • Built end-to-end data curation pipeline processing images, captions, and markdown files from 42,673 scientific articles, extracting and structuring 171K+ image-caption pairs and 165K+ figures vision-language model training. • Fine-tuned LLaVA vision-language models (13B parameters) using LoRA on 50,882 image-text pairs, with figure-caption linking. • Developed LLM-as-judge evaluation framework for assessing factual accuracy of multimodal model outputs against ground-truth scientific data. • Achieved 135% performance improvement and 96.6% hallucination reduction over baselines. • *Outcome: Oral presentation at IEEE ISBI 2025 (~20% acceptance rate)*
 - **Infrastructure & Collaboration:** • Utilized ALCF leadership-class HPC infrastructure for large-scale distributed multi-GPU model training and evaluation. • Collaborated with Argonne scientists in Computing, Environment and Life Sciences Directorate and X-ray Science Division. • Presented findings to DOE leadership, Argonne senior scientists; invited to Trillion Parameter Consortium (90+ institutions across 4 continents).
- **University of Alabama (Carnegie R1 University)** Tuscaloosa, AL
 - **Doctoral Researcher** — *AI for Infrastructure & Smart Cities* June 2021 – Present
 - **Real-Time Scientific Vision Pipeline:** • Designed and deployed end-to-end YOLO(v8/11-n,l,x) & RT-DETR-L/X pipeline for automated understanding of heterogeneous urban imagery across diverse geographic settings, sensor platforms, and environmental conditions. Built multimodal data ingestion, annotation, and training workflows producing production-ready models for emergency managers. Currently deploying **TornadoNet App** with VLM capabilities for active NIST use [PDF] (github.com/crumeike/TornadoNet-App). • *Outcome: 2nd place NSBE National Convention (15,000+ attendees), ASCE Editor's Choice Award.*

- **Large-Scale Architecture & Optimization Benchmark:** Designed **2,356 experiments** across **79 architectures** on curated QSTD dataset. Key findings includes that optimizer choice outweighs architecture. Notably SGD vs. Adam reversed ViT/Swin ranking with **+25–38 F1-point gains**. Champion model maintained high Ordinal Accuracy under zero-shot cross-event shift.
- **Multi-Institutional Research Leadership:** • Led technical coordination for NIST Disaster and Failure Studies Program consortium (NIST, Johns Hopkins, Colorado State, South Alabama). • Delivered bi-weekly progress updates to federal stakeholders. • Project showcased at NIST AI Day 2024.
- **IoT & Heterogeneous Data Integration:** • First application of OIIE international standards to real-time multi-modal urban data (sensor networks, geospatial, temporal streams). • Processed 25,000+ heterogeneous data points into structured decision-support representations. • *Outcome: 2 first-place innovation awards in 2020 & 2022, TISDIC 2023 first-authored publication.*

• **VFD Tech Limited (VFD Group Plc)**
IT Project Manager

Lagos, Nigeria
 June 2020 – June 2021

- **Product Management/Platform Development:** • Managed 25+ technology projects; led second major release of V mobile banking app (1M+ Android downloads). • Oversaw Banking-as-a-Service platform serving 20+ corporate clients.

EDUCATION

- **University of Alabama — Dissertation Defended & Passed** Tuscaloosa, AL
PhD, Civil Engineering (*AI/Computer Vision focus*) — *GPA: 3.85/4.0* Degree Conferred May 1, 2026
 – Dissertation: *An Interoperable Framework for Real-Time Infrastructure Monitoring and Deep Learning-Based Post-Disaster Damage Assessment for Urban Resilience*
- **University of Alabama** Tuscaloosa, AL
MS, Computer Science — *GPA: 3.85/4.0* Dec 2024
- **University of Alabama** Tuscaloosa, AL
MS, Civil Engineering — *GPA: 4.0/4.0* Dec 2023

KEY PROJECTS & OPEN SOURCE

- **PtychoBench** (github.com/crumeike/PtychoBench): Benchmark for evaluating foundation model adaptation to X-ray ptychography. Domain-aware VLMs and LLMs (Llama-3.1, Llama-3.2-vision). Published **NeurIPS 2025 AI4MAT**.
- **LUCID** (github.com/crumeike/lucid-biomedical-vlm): End-to-end pipeline: 42K article artifact ingestion, 171K image-caption curation, LoRA fine-tuning, LLM-as-judge evaluation. Published **IEEE ISBI 2025**.
- **TornadoNet** (github.com/crumeike/TornadoNet): Real-time building damage detection with ordinal supervision using YOLO/RT-DETR. Multi-institutional **NIST consortium project 2026**. [[TornadoNet App](#)]

SELECTED PUBLICATIONS (11 TOTAL, 8 FIRST-AUTHOR, 2 SOLE-AUTHOR)

- **R. Umeike**, C. Pham, R. Hausen, T. Dao et al., "TornadoNet: Real-Time Building Damage Detection with Ordinal Supervision," *arXiv preprint arXiv:2603.11557*, 2026. [[PDF](#)]
- **R. Umeike**, T. Dao, S. Crawford, J. van de Lindt et al., "Architectural Insights for Post-Tornado Damage Recognition," *arXiv preprint arXiv:2602.14523*, 2026. [[PDF](#)]
- **R. Umeike**, N. Getty, X. Yin, Y. Jiang, "Adapting General-Purpose Foundation Models for X-ray Ptychography in Low-Data Regimes," *NeurIPS 2025 AI4MAT Workshop*, 2025.
- **R. Umeike**, N. Getty, F. Xia, R. Stevens, "Scaling Large Vision-Language Models for Enhanced Multimodal Comprehension in Biomedical Image Analysis," *IEEE ISBI 2025 (Oral)*, 2025.
- **R. Umeike**, "Contact-Based Grasp Control and Inverse Kinematics for a Five-Fingered Robotic Hand," *IEEE ICMRE 2025*, pp. 116-121, 2025.
- **Full publication list:** scholar.google.com/citations?user=IeJsfl8AAAAJ

HONORS & RECOGNITION

- **NeurIPS AI4MAT Travel Grant** - Competitive award for conference presentation (2025)
- **2nd Place, NSBE National Technical Research Exhibition** - 15,000+ attendees (2025)
- **ASCE Editor's Choice Award** - Awarded to < 2% of all 200,000+ ASCE publications (2024)
- **Argonne Sustainable Research Pathways Fellowship** - Competitive fellowship for placement in DOE (2024)
- **Invited Speaker, Trillion Parameter Consortium** - Global AI consortium, 90+ institutions, 4 continents (2024)
- **1st Place, UA Faculty & Staff Innovation Competition** – Led technical implementation of e-Flood IIoT system; \$5,000 First Prize, Division of Finance and Operations (2022)
- **Media Features:** [Argonne National Laboratory](#), [ALCF](#), [TPC Seminar Series](#), [ASCE Civil Engineering Source](#)